

1. Administrative & Study Identification

Full Study Title: Exploratory Evaluation of Novel Investigational Eye Movement Biomarkers to Track Ofatumumab Treatment Response in Canadian Patients With Active Relapsing-Remitting Multiple Sclerosis (ELIOS) **Short Title/Acronym:** ELIOS - Investigational Biomarkers to Track Disease Modification in Active RRMS **Protocol Number:** COMB157GCA05 **NCT Number:** NCT06733922 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Ofatumumab and ETNA-ProgMS (Eye Tracking Neurological Assessment) SaMD **Phase of Development:** Phase 4 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the change in Eye Movement Biomarkers (EMB) to track disease-related modifications in patients with active RRMS. **Secondary Objectives:** To evaluate the change in EMB trajectories at 3, 6, 12, and 24 months, and to assess Patient-Reported Outcomes (PROs).

2.2 Background & Rationale This trial explores a novel method to track changes in multiple sclerosis (MS) using eye movements. It focuses on patients with relapsing-remitting multiple sclerosis (RRMS) who are initiating treatment with ofatumumab, an anti-CD20 immunotherapy. The study aims to determine if precise eye-tracking using the investigational software (ETNA-ProgMS) can provide new, reliable insights into how MS affects the central nervous system over time in a real-world setting.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional / Diagnostic Study **Control Type:** Single-arm (No placebo group) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 224 participants **Number of Sites:** Multiple sites across Canada **Estimated Study Start:** 27-Nov-2024 **Estimated Primary Completion:** 30-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients (\$\ge\$ 18\$ years) who are prescribed ofatumumab as part of routine clinical care but have not yet received their first dose. 2. Patients who meet the

Expanded Disability Status Scale (EDSS) score range of 0 up to 7 at the time of screening. 3. Confirmed diagnosis of active RRMS according to the 2017 Revised McDonald criteria.

4.2 Exclusion Criteria 1. Patients with primary progressive MS, secondary progressive MS without disease activity, clinically isolated syndrome, or radiologically isolated syndrome. 2. Prior exposure to anti-CD20 B-cell therapy (e.g., ocrelizumab, ofatumumab) or natalizumab; or treatment with cladribine/alemtuzumab within 12 months prior to baseline. 3. Pregnant or nursing (lactating) women, or women of child-bearing potential not utilizing stable, effective contraception.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ofatumumab administered as per standard of care routine (monthly after initial loading), coupled with regular precision eye-tracking assessments. **Route of Administration:** Subcutaneous (Ofatumumab) and Device-based software interface (ETNA-ProgMS SaMD v1.0.11 or later). **Duration of Treatment:** Up to 36 months of follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant symptom management medications as required for MS. **Prohibited:** Other disease-modifying therapies (DMTs) such as cladribine, alemtuzumab, natalizumab, and other anti-CD20 B-cell therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures	:---	:---
:---	Screening/Baseline	Day -14 to 0	Informed Consent,	
			EDSS Evaluation, MRI, Blood tests, Initial ETNA-ProgMS Tracking	
Treatment Phase	Month 0	First dose of ofatumumab initiation		
Interim	Months 3, 6, 12, 24	Safety Labs, MRI (per		
		standard of care), Eye Movement Biomarkers tracking, PRO		
assessment	End of Study	Month 36	Final EDSS, ETNA-ProgMS	
tracking, Exit Interview				

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change in Eye Movement Biomarkers (EMB) from baseline. **Measurement Method:** Eye Tracking Neurological Assessment (ETNA-ProgMS) Software as a Medical Device (SaMD).

7.2 Secondary & Exploratory Endpoints 1. Change in EMB trajectories at 3, 6, 12, and 24 months. 2. Patient-Reported Outcomes (PRO) regarding symptomatic burden and quality of life.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular clinical care visits, tracking infusion-related reactions, and monitoring device usability. **Serious Adverse Events (SAEs):** Standard reporting timelines within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Managed by Novartis Pharmaceuticals safety review board.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary outcome trajectories and Safety Set for safety metrics evaluating patients who receive ≥ 1 dose of ofatumumab. **Sample Size Justification:** $N=224$, powered to detect significant longitudinal changes in EMB patterns corresponding with disease stabilization. **Handling of Missing Data:** Standard longitudinal imputation techniques (e.g., Mixed-Effect Model Repeated Measure - MMRM) for device-captured biomarker data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Periodic site and remote monitoring to ensure clinical care compliance and SaMD software validation accuracy. **Audit Trail:** Secure eCRF locking and encrypted digital biomarker data capture from the ETNA-ProgMS platform.

11. Study Identifiers & Governance

Primary ID: COMB157GCA05 **Registry Number:** NCT06733922 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** ELIOS **Official Regulatory Title:** Exploratory Evaluation of Novel Investigational Eye Movement Biomarkers to Track Ofatumumab Treatment Response in Canadian Patients With Active Relapsing-Remitting Multiple Sclerosis (ELIOS)

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Relapsing-Remitting Multiple Sclerosis (RRMS) **Diagnosis Threshold:** 2017 Revised McDonald criteria, EDSS 0-7 **Medical Methodology:** Ofatumumab immunotherapy paired with digital biomarker tracking **Optimization Target:** Tracking disease-related modification and disability progression

13. Study Procedures & Methodology

Management Pathway: Standard clinical care for active RRMS
Education Protocol (HCP): ETNA-ProgMS SaMD device calibration and operations
Education Protocol (Patient): Self-administration of ofatumumab and device orientation
Quality Audit Mechanism: SaMD tracking accuracy, data synchronization, and patient testing adherence

14. Observation & Follow-Up Schedule

Total Duration: Up to 36 Months
Onsite Visit Cadence: Baseline, Months 3, 6, 12, 24, and 36 (or per standard of care)
Remote Monitoring Frequency: ETNA-ProgMS remote data capture
Checklist Review Interval: Every 3 to 6 months aligned with clinical assessments

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				